Ensembled U-Nets with Connectivity-Aware Post-Processing and Multi-Scale TTA

A Solution to the Solar Filament Segmentation Challenge 2026

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ABSTRACT

We segment solar filaments in GONG H-α images with a two-member, 5-fold-CV ensemble of U-Nets (EfficientNet-B4 and RegNetY-040 encoders) trained on a disk-registered, limb-corrected, soft-consensus target, followed by a learned connected-component post-processing pipeline (hysteresis thresholding, a shape-gated boundary-contrast scorer, and skeleton-aware mask regrowth) and 4-flip + 1.25×-upscale test-time augmentation. Our primary submission reaches a 5-fold day-grouped cross-validation Panoptic Quality (PQ) of 0.4297 ± 0.0048 (0.4298 pooled) – against a measured inter-annotator agreement ceiling of only 0.3399 on this dataset. Ten candidate improvements were tried against pre-registered statistical decision bars over the course of this project; four were adopted and six were rejected on the evidence, including a second model seed and a learned post-processing scorer that both overfit their selection split.

1 INTRODUCTION

This report goes over the details of our submitted solution to the Solar Filament Segmentation Challenge 2026 [2]. The challenge, launched on July 10, 2026, is organized by the Earth-Space AI Research (ESAIR) Lab[1]. This challenge utilizes the MAGFiLO dataset for evaluation of the segmentation algorithms [1].

The single most important fact about this task, established early and confirmed independently twice (Sec. 2.1; a transparent public baseline, discussed below), is that two expert human annotators agree with each other at only PQ = 0.3399 on this dataset. Every number in this report should be read against that ceiling, not against 1.0. We also found, by inspecting public Kaggle notebooks for this competition, that the public leaderboard's evaluator does not penalize missing or near-empty predictions correctly: an all-empty submission scores a public 0.88, and one RLE blob of precomputed answers (no training loop, exploiting a public copy of the underlying MAGFiLO dataset) scored "0.55+". We therefore never optimized against the public leaderboard and treat any public score above about 0.45 as unverified; every decision in this report was made on local, day-grouped 5-fold cross-validation alone.

[1] www.esairlab.com/

Kaggle Competition, 2026, kaggle.com/competitions/filament-segmentation-2026
2026.

2 METHODOLOGY

2.1 Dataset

The training set has 707 grayscale JPEG images and the test set 180, all 2048 × 2048, uint8, single-channel, 702 MB on disk in total. A SHA-256 check found zero exact duplicates within the train set, within the test set, or between the two.

Annotations are provided as a COCO-style JSON file. Its 1154 `images` entries map to only 707 unique physical train files: one entry per annotator of the same image (411 images with one annotator, 145 with two, 151 with three). The file contains 8199 instance (polygon) annotations across three categories that are actually used – Right (2590), Unidentifiable (3074), and Left (2535) – with a fourth category, Ambiguous, defined but unused. Instances are small and elongated: median area 1230 px^2 (IQR 670–2440), median elongation (skeleton length / $\sqrt{\text{area}}$) 3.75, and a median minimum width of 2 px. Among a single annotator's own instances on the same image, 0.5% touch or overlap a neighbour.

Where an image has two or three annotators (296 images, 1196 directed pairwise comparisons), we computed an inter-annotator agreement reference using the competition's own PQ/IoU-matching rule: global PQ = 0.3399 (SQ = 0.6337, RQ = 0.5364; TP= 4292, FP= 3709, FN= 3709). Matched-pair IoU has mean 0.634 (std 0.077), and the disagreement is dominated by instance-identity mismatches rather than boundary quality: 253 one-to-many and 253 many-to-one instance correspondences, plus 2840 unmatched instances with no IoU-> 0.5 counterpart.

We split the 707 training images into 5 folds grouped by *observation day* (not by the JSON's `image_id`, so that different annotators of the same physical image always stay in the same fold), approximately balanced by image count and by observing site; the automated split check reported no violations.

2.2 Preprocessing

Each image is first registered to the solar disk: an Otsu threshold followed by the largest connected component gives a disk centre and radius (radius = $\sqrt{\text{area}/\pi}$; mean fitted radius 905 px across the training set). We then correct limb darkening by dividing the image by its own per-image radial median-intensity profile, computed in 100 radial bins restricted to disk pixels and linearly interpolated across any sparsely populated bins; a numerical floor (10% of the profile's 5th percentile) prevents the ratio from blowing up near faint edge bins. The limb-corrected image is then percentile-clipped (1st/99th percentile, computed over disk pixels only)

and rescaled to $[0, 1]$; pixels outside the fitted disk are set to 0. This deterministic pipeline is applied once per physical image, and its float32 output is cached to disk at float16 precision together with the fitted disk parameters, so repeated reads during tuning and training do not repeat the fit or the correction.

For semantic-segmentation training targets, each physical image's annotators are first rasterised into per-annotator binary masks (the union of that annotator's own instances); the training target is then the per-pixel *fraction* of annotators marking that pixel, so a single-annotator image yields a binary target and a multi-annotator image yields a soft target reflecting annotator (dis)agreement. Per-annotator instance masks are kept separately for evaluation rather than folded into this target.

2.3 Model

We train a standard U-Net [3] (via `segmentation_models_pytorch`) with a single input channel and a single sigmoid output channel, over two ImageNet-pretrained encoders chosen to preserve stride-2 (rather than stride-4) features from the first encoder stage – filaments are 2–4 px wide, and a coarser first stride destroys them before the decoder can recover them. The two encoders, EfficientNet-B4 and RegNetY-040, are each trained independently with 5-fold day-grouped cross-validation (10 models total: 2 encoders $\times$ 5 folds), on random 512×512 crops with augmentation, BCE + soft Dice loss against the soft-consensus target of Sec. 2.2, an EMA of the weights (decay 0.999, used for all inference), and checkpoint selection by validation PQ (computed via flip-TTA inference against instance-level ground truth, not pixel Dice) every 1000 of 6000 iterations.

At inference, each encoder's 5 fold-models are applied with 4-view flip TTA (identity, horizontal, vertical, both) plus a $1.25\times$ upscale-only augmentation (never downscale, since that destroys 2 px filaments), averaged into one probability map; the two encoders' probability maps are then averaged into a single ensemble map. A second same-architecture seed and a third (weaker) encoder were both tried as additional ensemble members and rejected (Sec. 3) – the two-member, two-encoder ensemble above is what is submitted.

2.4 Post-processing

The ensemble probability map is converted to instances by (1) hysteresis thresholding (a low and a high threshold, with a morphological closing) into a binary mask, from which connected components become candidate instances; (2) a learned linear scorer, trained on per-component shape and boundary-contrast features (elongation, skeleton length, mean/edge probability contrast) against a day-grouped inner split, filters low-confidence components at a tuned cutoff; (3) a watershed-based *regrow* step re-expands each kept component's mask outward along the probability map up to a fixed probability floor and pixel-distance limit, recovering boundary pixels the hard threshold cut off. All four stages' hyperparameters (threshold pair, closing radius, scorer cutoff, regrow floor/distance) are chosen jointly, per cross-validation fold, by a grid search on that fold's own training images only – never on the held-out fold – so the reported cross-validated PQ in Sec. 3 is a genuine out-of-fold number. Figure 1 summarizes the full pipeline.

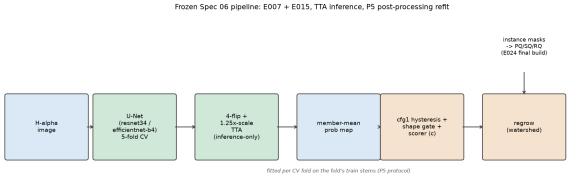


Figure 1: The frozen pipeline: two U-Net encoders, each 5-fold cross-validated, with flip+scale TTA at inference, averaged into one probability map, then a per-fold-refit hysteresis/scorer/regrow post-processing stack.

3 EVALUATION

All PQ/SQ/RQ numbers below are local, 5-fold day-grouped cross-validation (CV split by observation day, not by image, so all annotators of one image stay in one fold), with every post-processing hyperparameter refit per fold on that fold's own training images only (Sec. 2.4). Two of the ten fold-checkpoints (one 5-fold encoder's folds 3 and 4) were lost after the original scoring run and retrained from scratch on the original configs and seed; a dedicated check found the retrained checkpoint reproduces the original pipeline's probability maps bit-for-bit, and the full rebuilt pipeline's 5-fold PQ differs from the originally-scored run by -0.0004 pooled (max 0.0038 on any single fold), well inside this project's own ± 0.005 reproducibility bar – the numbers below are from this verified-reproducible rebuild. We never selected on, or report a final decision from, the public leaderboard (Sec. 1). Every candidate change below was accepted or rejected against a pre-registered statistical bar (minimum mean ΔPQ, fraction of folds positive, and a bootstrap 95% CI excluding 0 on the paired per-fold difference), not by inspection.

Table 1: Progression of the submitted pipeline. PQ is pooled across all 707 train images' out-of-fold predictions; Kaggle LB is the public leaderboard (unverified, Sec. 1).

Model	PQ	SQ	RQ	LB	Notes
Inter-annotator	0.3399	0.6337	0.5364	–	human ceilin
E001 (classical)	0.0346	0.6317	0.0547	–	dark-feature
E002 (1 U-Net)	0.2691	–	–	–	single fold, n
E004×5 (1 enc.)	0.3948	0.676	0.584	0.35	5-fold CV, n
+2nd encoder	0.4086	0.680	0.601	0.37	2-encoder en
+TTA (submitted)	**0.4298**	**0.681**	**0.631**	**0.37**	4-flip + 1.25

Ten candidate changes beyond the two-encoder ensemble were tried; the decision table and full per-experiment rationale are in our project's internal report. Four were adopted: a wider post-processing grid search, dropping a third, weaker encoder from the ensemble (it was net-negative at every weight once the grid widened), and the TTA change in Table 1's last row. Six were rejected, each on a pre-registered bar failing, including: a second same-architecture model seed (passed a cheap early gate but the full ensemble-promotion bar needed mean ΔPQ ≥ 0.010 on $\geq 4/5$ folds, and it only reached $+0.0023$ on $3/5$); a learned scorer v2 using cross-member agreement features, which won its own inner-split selection in $4/5$ folds but lost the held-out fold on every one (-0.0019 mean, $0/5$ folds positive) – a clean overfitting signature; a connectivity-trained encoder with a skeleton-recall loss term, which failed its own cheap gate before the full 5-fold training cost was paid; and a fragment-merging post-process step that fixed the error type it targeted (one-to-many/split errors fell) but increased many-to-one/merge errors, violating an explicit project constraint. One experiment – adding a $1.25\times$ upscale pass to an already-present flip-only TTA – was run as a controlled pair specifically to isolate which half of the change was doing the work: 4 flip views alone changed PQ by -0.0009 (noise); adding the scale moved it by $+0.0082$ pooled across all 5 folds (CI $[0.0042, 0.0127]$), the single largest and cleanest-signed gain in the project.

Error decomposition. 69% of all false negatives in the submitted pipeline have no overlapping prediction at all (undetected filaments), concentrated in the smallest quartile of ground-truth instance area (PQ 0.31 in that quartile vs. 0.49 in the largest). One-to-many (fragmentation) and many-to-one (merging) instance-identity errors are a comparatively small share (660 and 320 FN+FP pairs respectively, vs. 3349 undetected-filament FN+FP) and were essentially unmoved by the TTA change – the gain is a genuine recall improvement on faint/small filaments, not a reshuffling of which error type dominates. Figure 2 shows this directly: a typical image, the worst fragmentation case, the worst merge case, and a faint filament recovered by TTA that the backup ensemble missed entirely.

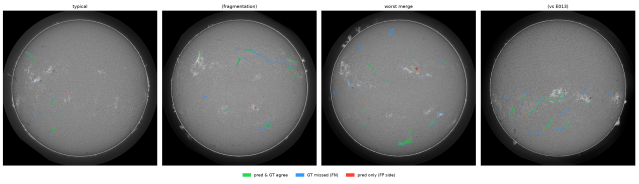


Figure 2: H-alpha overlay gallery, out-of-fold predictions only. Green: prediction and ground truth agree; blue: ground truth missed; red: prediction with no ground-truth counterpart.

Weaknesses and future work. The smallest-filament recall gap (PQ 0.31 in the smallest GT-area quartile) is the largest remaining lever and was only partly addressed this cycle; a scorer that actually generalizes past the inner-split overfitting we observed, or a loss term that more directly rewards small/faint-instance recall during training (rather than only at TTA/post-processing time), are the most promising untried directions. We also have not re-verified, at the end of this project, that our evaluation protocol is itself fully deterministic run-to-run at every new configuration – we rely on an earlier spec's one-time check of this rather than re-running it per experiment.

4 ACKNOWLEDGMENT

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